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Section: F

Activity: Genetic Algorithm

For Sections: B, E, F

Background

The **n-queens problem** involves placing n queens on an $n \times n$ chessboard such that no two queens attack each other. To adapt this to a Genetic Algorithm, we represent the board as a *chromosome*: an array where the index is the column and the value is the row position (1–8).

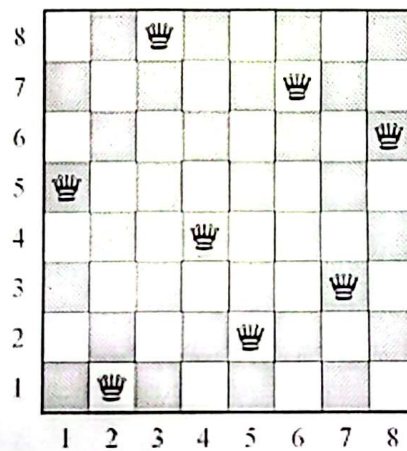


Figure 1: Example Chromosome: [5, 1, 8, 4, 2, 7, 3, 6]

Part 1 :

Consider the following initial population of four parents:

- P1: [2, 4, 1, 8, 3, 6, 7, 5]
- P2: [1, 5, 2, 6, 3, 7, 4, 8]
- P3: [4, 2, 7, 1, 8, 5, 3, 6]
- P4: [8, 6, 4, 2, 7, 5, 3, 1]

✕ 1.1 Fitness Evaluation

The fitness $f(x)$ is the number of **non-attacking pairs**. (Max fitness = 28).

$$f(P1) = \underline{25} \quad f(P2) = \underline{27}$$

$$f(P3) = \underline{25} \quad f(P4) = \underline{27}$$

X 1.2 Crossover

Perform **Single-Point Crossover** after the specified split points.

Constraint: This problem requires a *permutation* (each row 1-8 must appear exactly once). If an offspring has duplicate values, **repair it** by replacing all duplicates in the second half of the chromosome with the missing column numbers (in ascending order).

1. Offspring A (P1 and P2, split after index 3 (column 4)): [2, 4, 1, 8, 3, 7, 5, 6]
2. Offspring B (P2 and P1, split after index 3 (column 4)): [1, 5, 2, 6, 3, 4, 7, 8]
3. Offspring C (P3 and P4, split after index 5 (column 6)): [4, 2, 7, 1, 8, 5, 3, 6]
4. Offspring D (P4 and P3, split after index 5 (column 6)): [8, 6, 4, 2, 7, 5, 3, 1]

X 1.3 Mutation

Apply a random mutation to **Offspring A** and **Offspring C**:

- Mutated A (Change index 6 to value 2): [2, 4, 1, 8, 3, 7, 2, 6]
- Mutated C (Change index 0 to value 1): [1, 2, 7, 1, 8, 5, 3, 6]

Part 2: Conceptual Logic

- X 1. If the entire population has a "1" at index 0 and the mutation rate is 0%, can the algorithm ever place a queen in Row 5 for Column 1? Explain.
- X 2. In Task 1.2, we used a **Repair Function** to maintain the constraint.
 - X • What would happen to the fitness score if we left the duplicates?
 - X • If there is no repair function, how does this affect the search space (does it make it larger or smaller)?

1- No, it can not. Because the crossover swaps existing genes b/w the parents. So, even if we swap, we always get a 1 in column 1. We will need to mutate and introduce some new number at index 0 to place a queen in Row 5, Column 1.

2. • If we left duplicates, there will be more than 1 queens in a row and the fitness score might reduce.

- It makes the search space larger. The queens will be able to exist in same rows resulting in the requirement to scan rows for queen-queen collision as well.

Rough work

1.1:

8	0							
7				0				
6		0						
5					0			
4			0					
3						0		
2				0				
1								0
	1	2	3	4	5	6	7	8

Conflicts

f(A)

P1: 8-6, 7-6, 3-1 (25)

P2: 8-1 27

P3: 8-4, 7-3, 4-1 25

P4: 8-1 27

1.2: Crossover

OA: 2 4 1 8 3 7 ~~4~~ ~~8~~ ^{5 6}

OB: 1 5 2 6 3 ~~6~~ 7 ~~8~~ ^{4 8}

OC: 4 2 7 1 8 5 3 ~~8~~ ⁶

OD: 8 6 4 2 ~~7~~ 5 3 ~~6~~ ¹

1.3 Mutation:

M-A: 2 4 1 8 3 7 ~~2~~ ⁵ 6

M-C: (on Paper)

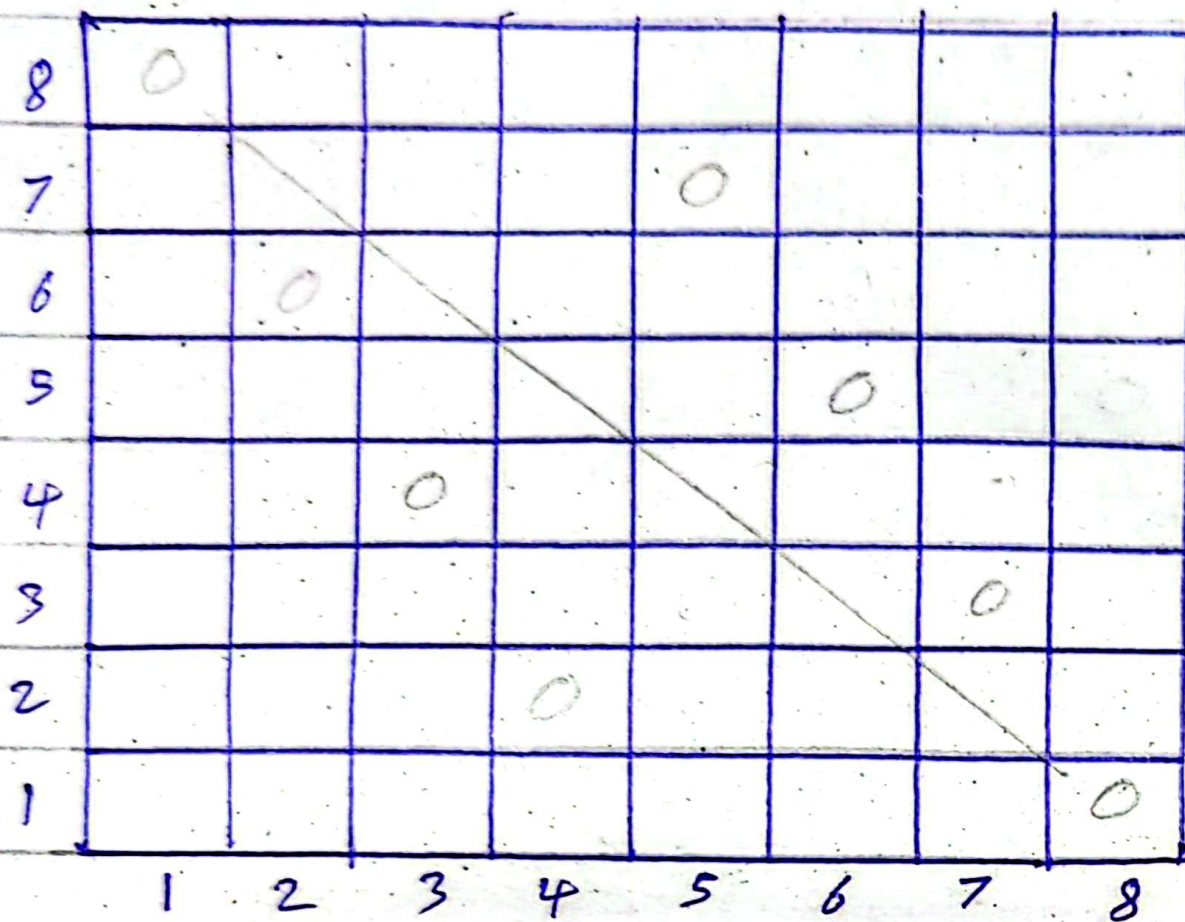
8								0
7						0		
6				0				
5		0						0
4							0	
3					0			
2			0					
1	0							
	1	2	3	4	5	6	7	8

P1: 8-6, 7-6, 3-1

P2: 8-1

P3:

f(6)
(25)
27



Conflicts	$f(A)$
P1: 8-6, 7-6, 3-1	(25)
P2: 8-1	27
P3: 8-4, 7-3, 4-1	25
P4: 8-1	27